

WEICHSELBERGER CHANNEL MODEL ANALYSIS REPORT

1. Objective

The objective of this work is to analyze the performance of the Weichselberger MIMO channel model under different coupling structures and evaluate:

- Effective Rank
- Singular Vector Coherence
- Required Pilot Ratio
- Required Observations
- NMSE (Normalized Mean Square Error)

The study uses matrix completion through the Singular Value Thresholding (SVT) algorithm to determine the minimum pilot overhead required to achieve a target NMSE of -15 dB.

2. System Parameters

Parameter	Value
Number of Receive Antennas (Nr)	16
Number of Transmit Antennas (Nt)	8
Number of Simulations	10
Target NMSE	-15 dB
Transmit Correlation Factor	0.7
Receive Correlation Factor	0.6

3. Procedure of the Weichselberger Model

Step 1: Generate Correlation Matrices

Transmit and receive correlation matrices are generated using exponential correlation:

$$R_t(m,n) = \rho_t^{|m-n|}$$

$$R_r(m,n) = \rho_r^{|m-n|}$$

where:

- $\rho_t = 0.7$
- $\rho_r = 0.6$

These matrices model spatial correlation among antennas.

Step 2: Eigenvalue Decomposition

The correlation matrices are decomposed as:

$$R_t = U_t \Lambda_t U_t^H$$

$$R_r = U_r \Lambda_r U_r^H$$

The eigenvectors represent dominant spatial directions.

Step 3: Construct Coupling Matrix $\tilde{\Omega}$

The coupling matrix $\tilde{\Omega}$ determines how energy is distributed between transmit and receive eigenmodes.

Different $\tilde{\Omega}$ structures create the four channel scenarios.

Step 4: Normalize Coupling Matrix

The matrix is normalized to maintain equal channel power:

$$\tilde{\Omega} = \tilde{\Omega} / \|\tilde{\Omega}\|_F \times \sqrt{N_r N_t}$$

Step 5: Generate Random Scattering Matrix

A complex Gaussian matrix G is generated:

$$G \sim \text{CN}(0,1)$$

representing independent scattering components.

Step 6: Generate Weichselberger Channel

The channel matrix is formed as:

$$H = U_r (\tilde{\Omega} \odot G) U_t^H$$

where:

- $\odot$ denotes element-wise multiplication.
 - $\tilde{\Omega}$ controls power coupling.
 - U_r and U_t provide spatial correlation.
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Step 7: Compute Effective Rank

Singular Value Decomposition is applied:

$$H = USV^H$$

Effective Rank:

$$r_{\text{eff}} = \exp(-\sum p_i \ln(p_i))$$

where:

$$p_i = \sigma_i / \sum \sigma_i$$

A smaller rank indicates stronger low-rank structure.

Step 8: Compute Coherence

Singular vector coherence is calculated to measure concentration of singular vectors.

Lower coherence generally improves matrix completion performance.

Step 9: Matrix Completion

Random pilot observations are selected using a binary sampling mask.

The SVT algorithm reconstructs the channel matrix.

Step 10: Binary Search for Minimum Pilot Ratio

The pilot ratio is adjusted through binary search until:

$$\text{NMSE} \leq -15 \text{ dB}$$

with at least 70% successful reconstructions.

The resulting pilot ratio and observations are recorded.

4. Description of the Four Weichselberger Cases

Case 1: Exponential Diagonal Decay

Definition

$$\tilde{\Omega}(r,t) = 0.9^{|r-t|}$$

Characteristics

- Energy concentrated near the main diagonal.
- Power decreases gradually away from the diagonal.
- Represents smooth eigenmode coupling.
- Preserves sparsity while allowing limited energy leakage.

Expected Behaviour

- Low effective rank.
 - Low coherence.
 - Good matrix completion performance.
 - Lower pilot requirement.
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Case 2: Block / Clustered Energy Distribution

Definition

High energy is assigned to specific blocks while remaining entries have very low energy.

Main blocks:

- Upper-left block $\rightarrow 0.8$
- Lower-right block $\rightarrow 0.6$
- Remaining entries $\rightarrow 0.05$

Characteristics

- Simulates clustered scattering environments.
- Energy concentrated in a few angular regions.
- Strong dominant propagation clusters.

Expected Behaviour

- Lowest effective rank.
 - Strongest low-rank property.
 - Excellent reconstruction capability.
 - Minimum pilot overhead among all cases.
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Case 3: Exponential Cross-Diagonal Decay

Definition

$$\tilde{\Omega}(r,t) = 0.9^{|r-(Nt-t+1)|}$$

Characteristics

- Energy concentrated along the anti-diagonal.
- Represents coupling between opposite eigenmodes.
- Smooth version of anti-diagonal sparsity.

Expected Behaviour

- Moderate effective rank.
 - Increased coherence.
 - More difficult matrix completion.
 - Higher pilot requirement compared to Cases 1 and 2.
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Case 4: Banded Diagonal Coupling

Definition

$$\tilde{\Omega}(r,t) = 0.5, \text{ if } |r-t| \leq 2$$

$$\tilde{\Omega}(r,t) = 0.05, \text{ otherwise}$$

Characteristics

- Thick diagonal band.
- Energy spread over several neighboring modes.
- Less sparse than Cases 1 and 2.
- Richer multipath coupling.

Expected Behaviour

- Highest effective rank.
 - Higher coherence.
 - Largest pilot requirement.
 - Most difficult reconstruction scenario.
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5. Comparative Summary

Case	Energy Structure	Rank	Coherence	Pilot Requirement
Case 1	Exponential Diagonal	Low	Low	Low
Case 2	Block/Clustered	Very Low	Very Low	Lowest

Case	Energy Structure	Rank	Coherence	Pilot Requirement
Case 3	Cross-Diagonal	Medium	Medium	Medium
Case 4	Banded Diagonal	Highest	Highest	Highest

6. Conclusion

The Weichselberger model provides a flexible framework for modeling realistic MIMO channels by explicitly controlling eigenmode coupling through the coupling matrix $\tilde{\Omega}$.

Among the four scenarios:

- Case 2 produces the strongest low-rank structure and requires the least pilot overhead.
- Case 1 also performs well due to smooth diagonal energy concentration.
- Case 3 introduces anti-diagonal coupling, increasing reconstruction difficulty.
- Case 4 distributes energy across a wider diagonal band, resulting in the highest rank and largest pilot requirement.

Therefore, channel sparsity and coupling structure have a significant impact on effective rank, coherence, and matrix completion performance.